

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018- 2019 (Odd Semester)

Innovative Practices Description

UNIT I (INTRODUCTION TO OPTICAL FIBERS)

Degree, Semester& Branch: VII Semester B.E. ECE A & B

Course Code & Title: EC6702 & Optical Communication and Networks

Name of the Faculty member: Mr.P.Gunasekaran

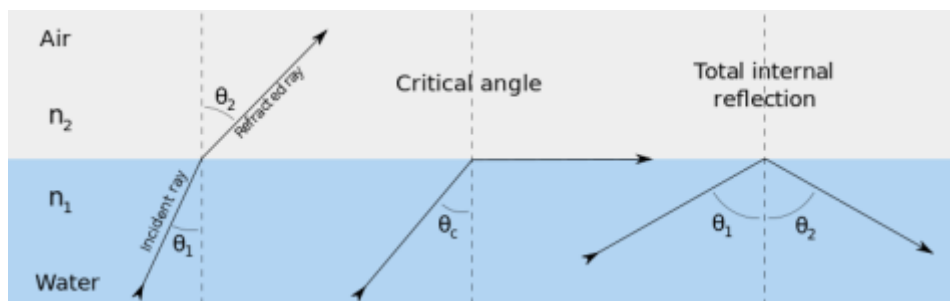
Name of the Topic: Total Internal Reflection

Name of the Innovative Practice: Animation video

Description:

Total internal reflection refers to the complete reflection of a ray of light within an optically-denser medium from the surrounding surfaces of optically less dense media back into the denser medium.

Light ray travel from an optically denser medium to a less dense medium. The angle of incidence must be greater than a certain angle, called the critical angle.



For a ray of light passing from an optically denser to a less dense medium, critical angle, c , is the angle of incidence at which the angle of refraction is 90° .

When the angle of incidence is less than the critical angle, the ray passes out into the less dense medium. When the angle of incidence is greater than the critical angle, the ray is reflected back into the denser medium.

References:

1. <https://www.youtube.com/watch?v=T-iy29oEmSI>
2. <https://www.khanacademy.org/science/physics/geometric-optics/reflection-refraction/v/total-internal-reflection>